

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025–present	Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea Deep-learning multimodal-integration models fusing imaging, transcriptomic, and clinical data into single predictive models.
Associate Director 2023–2025	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Led a humanized anti-CD40 antibody program: candidate ranking by surface plasmon resonance (Biacore) and ELISA, receptor-blocking confirmation, human and non-human-primate cross-reactivity, in vivo efficacy in primates, and the regulatory documentation. Placed and oversaw CRO and core-facility work, from study design and assay conditions through data delivery and go/no-go calls. Built the in-house single-cell and spatial analysis pipeline. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs.
Manager 2021–2023	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Wrote the two KDDF proposals that funded the anti-CD40 program and ran the lead-selection work under them.
Research Student 2020–2023	SCAID, Single Cell Atlas of Immune Diseases, Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea Contributed the founding proposal; executed within the funded program (700+ donors × 8 organs): milestone tracking, multi-site QC, multi-PI coordination.
Researcher 2019–2021	Xenotransplantation Research Center, Seoul National University, Seoul, Republic of Korea Rhesus macaque tissue processing and longitudinal immune monitoring across transplantation studies.
Teaching Assistant 2014–2016	Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

PUBLICATIONS

18 TOTAL PAPERS	17 PEER-REVIEWED	1 IN REVISION	6 FIRST / CO-FIRST
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Google Scholar · Full Profile, Citations & h-index ↗

PUBLICATIONS · FIRST / CO-FIRST AUTHOR

2026	Woogil Song*, Seong-Jun Kang* , Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." <i>Nature Immunology</i> . in revision
2025	Seong-Jun Kang* , Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8 ⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." <i>Journal of Neuroimmunology</i> . Download PDF
2025	Seong-Jun Kang* , Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." <i>Allergy</i> . Download PDF <i>Co-first author. Single-cell and functional dissection of skin-homing regulatory T cells in atopic dermatitis, with OX40 up-regulated alongside impaired suppressive function.</i>

2024	Jong-Min Kim*, Seong-Jun Kang* , So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." Xenotransplantation . Download PDF Co-first author. Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques, with longitudinal immune monitoring.
2022	Youngkyoung Lim*, Seong-Jun Kang* , Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." JEADV . Download PDF
2019	Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, Seong-Jun Kang* , Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." Islets . Download PDF Co-first author, one of four authors contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

CONTRIBUTING AUTHOR

2026	Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, Seong-Jun Kang , Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." Allergy .
2024	Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, Seong-Jun Kang , Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." eBioMedicine . Download PDF
2024	Youngkyoung Lim*, Beom Keun Cho*, Seong-Jun Kang , Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus." JEADV . Download PDF
2023	Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORγt through posttranslational modification." Experimental & Molecular Medicine .
2022	Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, Seong-Jun Kang , Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation." Experimental & Molecular Medicine .
2021	So Hee Hong, Hyun Je Kim, Seong-Jun Kang , Chung-Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." Current Diabetes Reports . Download PDF
2020	Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." Immunology Letters .
2019	Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bonggi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, Seong-Jun Kang , Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation." Scientific Reports .
2018	Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, Seong-Jun Kang , Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model." Xenotransplantation .

- 2018** Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." *Xenotransplantation*.
- 2018** Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, **Seong-Jun Kang**, Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." *Transplantation*.
- 2018** Jung-Sik Kim, Hyunwoo Chung, Nari Byun, **Seong-Jun Kang**, Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." *Biochemical and Biophysical Research Communications*.

MANUSCRIPTS IN PREPARATION

- 2026** Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." in preparation
- Co-author (immunology). Ran the Olink plasma proteomics across 12 patients and 6 controls and the flow-cytometry immunophenotyping of lymph-node leukocytes.*

TECHNICAL EXPERTISE

Multiplexed in situ detection and imaging	GeoMx DSP run end to end — slide preparation, oligonucleotide-barcoded probe hybridisation, immunofluorescence staining, fluorescence-guided ROI selection, instrument operation, UV cleavage and barcode collection. MACSima cyclic multiplexed immunofluorescence, iterative stain-image-erase rounds on one section. Olink targeted plasma proteomics; Visium spatial transcriptomics. High-parameter fluorescence panel design up to 24 markers. Spinning-disk confocal (Nikon CSU-W1 SORA) with quantitative image analysis.
High-content image and data analysis	Segmentation, feature extraction and model scoring on public iPSC-neuron imaging screens. Batch- and donor-aware modelling, held-out evaluation, calibrated confidence. Automated library preparation on 10x Chromium Connect .
Cell, tissue & immunological assays	Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. Olink targeted plasma proteomics (immune & neurology panels).
Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5'/3' library generation and 10x Chromium Connect automated preparation. Seurat , Scanpy , Squidpy , Monocle3 , CellChat/NicheNet ; scVI/scANVI, batch integration, consensus NMF ; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: GeoMx DSP run end to end, Visium , MACSima cyclic multiplexed immunofluorescence.
Computational	Python: scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch ; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.

Microscopy & animal models	Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.
Program & external execution	CRO and core-facility work from study design and placement through data delivery and go/no-go calls; assay conditions specified to the vendor; regulatory documentation for a therapeutic antibody program; funder milestones and reporting; multi-site sample QC across a 700-donor collection.

BOOKS

Seong-Jun Kang, Do-gyun Kim, Jin-seon Han. **AI Drug Discovery: A Practical Guide for Medical and Biomedical Researchers** (title translated). Beomun Education. Forthcoming September 21, 2026. 548 pages. ISBN 979-11-5943-583-6. [Publisher](#)

The book covers molecular and protein data preparation, QSAR, ChemBERTa, graph neural networks, docking, virtual screening, candidate optimization, and follow-up experimental validation.

SELECTED GRANT WRITING & PROGRAM FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER → ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Anti-CD40 antibody lead selection and optimization (multiple sclerosis)	2021–2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Korea Drug Development Fund (KDDF)	Anti-CD40 lead compound (PB101): efficacy and pharmacodynamic biomarkers (pemphigus vulgaris)	2022–2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	~\$222,000	Lead proposal writer; managed >90% of execution
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher
NAVER Corporation	Engineered regulatory T cells for immune-mediated diseases (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT)	Multi-layered immune-disease single-cell map for molecular target discovery (SCAID, A9-22-1-05)	2022–2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020–2021	~\$40,000	Grant writing, experimental design & technical validation

*All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, **not as PI**. Budgets converted at ≈ ₩1,400 / USD.*

■ AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

- 2024** **Antagonistic Anti-CD40 Humanized Antibody.** Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.
- 2019** **Immunosuppression Composition Comprising JAK Inhibitor.** Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20.

AWARDS

- 2024** **Grand Prize**, 2024 GPTers AI Hackathon (Project GAIA Team)
- 2023** **Grand Prize**, SBA Smart Workathon (Work Process Innovation)
- 2017** **Best Poster Award**, American Transplant Congress, Chicago · **Best Presentation Award**, Korean Association of Immunologists

SCHOLARSHIPS

- 2014-2020** NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship

■ SELECTED PRESENTATIONS

- Poster** KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)

■ MENTORING AND RESEARCH TRAINING

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| <p>2013 Sung Ha Lim, M.D., Ph.D.
STUDENT INTERN
Postdoctoral Fellow; Laboratory of Shadmehr Demehri, M.D., Ph.D.; Department of Dermatology, Massachusetts General Hospital; Harvard Medical School, Boston, MA, USA
Training: Cell culture, Western blotting, flow cytometry, and C57BL/6 mouse breeding.</p> | <p>2015 Soyoung Jeong, Ph.D.
STUDENT INTERN
Postdoctoral associate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea
Training: Cell culture, flow cytometry, and in vivo mouse experiments.</p> |
| <p>2016 Sowon Choi, M.S.
INCOMING GRADUATE STUDENT
Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea
Training: Mouse pancreatic islet isolation, pancreatic islet beta-cell functional assays, and ion channel assays.</p> | <p>2021 Hyo Jeong Nam, M.S.
INCOMING GRADUATE STUDENT
Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea; Visiting Scholar; Laboratory of Seunghee Kim-Schulze, Ph.D.; Department of Immunology and Immunotherapy; Icahn School of Medicine at Mount Sinai, New York, NY, USA
Training: GeoMx DSP, T-cell functional assays, and single-cell RNA sequencing (scRNA-seq) data analysis using R.</p> |
| <p>2021 Beom Keun Cho, M.S.
INCOMING GRADUATE STUDENT
Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea
Training: GeoMx DSP, mouse breeding, cell culture, T-cell functional assays, and scRNA-seq data analysis using R.</p> | <p>2021 Minji Kang, Ph.D.
INCOMING GRADUATE STUDENT
Postdoctoral associate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea
Training: GeoMx DSP spatial transcriptomics data analysis and T-cell functional assays.</p> |

2021	Jung Ho Lee, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate, Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea; Visiting Scholar, Laboratory of Ulrich von Andrian, M.D., Ph.D.; Department of Immunology, Harvard Medical School, Boston, MA, USA Training: Analysis of scRNA-seq data using R.	2021	Hyun Seung Choi, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Analysis of scRNA-seq data using R.
2022	Yuji Lee, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.	2022	Sanha Hwang, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.
2022	Dongjun Kim, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.	2022	Yewon Moon, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.
2022	Karoliina Eliisa Ruhnö, M.S. INCOMING GRADUATE STUDENT Senior Researcher; IMBiologics Corp., Suwon, Republic of Korea Training: Regulatory T-cell suppression assays.	2023	Moongi Chae, M.S. STUDENT INTERN Ph.D. Student; Johns Hopkins University, Baltimore, MD, USA Training: Immune-cell culture, flow cytometry, and scRNA-seq data analysis using machine-learning models and AI tools.
2023	Sung Bin Bae, M.S. INCOMING GRADUATE STUDENT Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Cell culture, flow cytometry, and GeoMx DSP spatial-transcriptomics data analysis.	2024	Mi Jeong Oh, M.S. MANAGER, PB IMMUNE THERAPEUTICS Manager; PB Immune Therapeutics, Seoul, Republic of Korea Training: Experimental design for animal sample collection, ELISA and SPR assay design, and 10× Genomics scRNA-seq.
2024	Eun Sol Jo, B.S. ASSISTANT MANAGER, PB IMMUNE THERAPEUTICS Assistant Manager; PB Immune Therapeutics, Seoul, Republic of Korea Training: Animal experiment design and evaluation of a mouse model of experimental autoimmune encephalomyelitis (EAE).		

REFERENCES

Chung-Gyu Park, M.D., Ph.D.

Professor, Microbiology & Immunology / Biomedical Sciences, SNU College of Medicine; CEO/CSO, PB Immune Therapeutics · chgpark@snu.ac.kr

Doctoral advisor and industry supervisor; directed the anti-CD40 antibody program.

Seung-Kuy Cha, Ph.D.

Professor, Dept. of Physiology & Global Medical Science, Yonsei University Wonju College of Medicine · skcha@yonsei.ac.kr

Most recent research supervisor (2025–2026).

Hyun Je Kim, M.D., Ph.D.

Assistant Professor, Microbiology & Immunology / Biomedical Sciences; Interdisciplinary Program in Artificial Intelligence (IPAI), SNU · tte9801@snu.ac.kr

Co-corresponding on the single-cell studies; supervised the computational work.

Jeong Seok Lee, M.D., Ph.D.

Associate Professor, Graduate School of Medical Science & Engineering, KAIST, Daejeon · chemami@kaist.ac.kr

Corresponding author, PDE4D / Nature Immunology study.